

Distributed agile: project management in a global environment

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Abstract Agile methods have been gaining acceptance in the mainstream software development community. At the same time, globally distributed software development is another trend delivering high-quality software to global users at lower costs. Little is published about the adoption and adaption of Agile methods in a distributed team and software globalization/localization project environment. The overall performance and satisfaction with the international deployment of the latest version of My Yahoo! increased by more than 30% after the global product team, distributed over three continents, adopted Agile methods. Our objective is to highlight successful practices and challenges that have been overcome by the globalization project, and suggest a framework for software globalization project management using a distributed Agile approach.

Keywords Agile methods · Scrum · Distributed software development · Software globalization

1 Introduction

Software globalization projects often meet not only internal challenges but also external challenges such as tight, shifting schedules, scope creep, inflexible budgets, and unrealistic customer expectations. These challenges typically result from a lack of process awareness and support on the part of development, sales, and management divisions (Flarup 2007). With software development going global, globally distributed projects are increasingly becoming the norm for large software systems. While this is a trend in the IT industry, there are also many organizations which now realize that traditional heavy-weight processes are

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slow, and lack interaction between customers and the product development teams (Schwaber 2008; Salo and Abrahamsson 2008; Dybå and Dingsøyr 2008).

The Global My Yahoo! group experienced the same aforementioned challenges, when they released 17 international versions of the old My Yahoo! ‘Zorro’ between 2005 and 2006. The product development team in Sunnyvale, California provided the international teams with a poorly internationalized Zorro platform and a huge international deployment manual along with a deployment checklist. The international teams were distributed across Asia Pacific, Europe and the Americas and were responsible for localizing the global product. The product development team was distributed in Sunnyvale and Bangalore, and supported the localization work mostly online. Each localization project took between 6–12 months and depended on a number of local modules and resources in rolling out each localized version of Zorro. As a result, the international deployment project for Zorro was under-resourced and under supported and didn’t work efficiently after all the effort put into it. This is where Agile methods will prove to be a benefit.

The global product team in Sunnyvale had adopted Scrum and Agile in 2006. The global product team and international teams released the next versions of My Yahoo! ‘Chameleon’ for the US and 17 international countries between 2007 and 2008. A great improvement has been made in both the process and the product quality. The Chameleon platform has strengthened the function of local configuration and programming beyond the new fancy features. Also, the overall performance and satisfaction with the international deployment of Chameleon compared to Zorro increased by more than 30%, even though only some Agile practices were followed.

This experience report presents an industrial case study that shows successfully integrated practices from agile software development (ASD) and distributed software development (DSD). The paper is organized as follows: we describe the study method in Section 2 and present the case study in Section 3. Then, we present the findings from the study and discuss the results in Section 4, and offer our conclusions in Section 5.

2 Study Method

In order for the case study results to have meaning, it is necessary to record contextual information and to implement measures that satisfy the goals of the study. One of the authors participated in the international deployment projects of both Zorro and Chameleon from the beginning to the end, and this case study is based on that individual’s experience as an international program manager who led the agile movement in a global context. Data collection for this study was largely dependent on historical data. The project planning/tracking tool and software source code control system were used for quality and productivity analysis, and the web-based defect/task tracking system provided a means of gathering quality information. In order to solicit qualitative information, we also had frequent communication with the global product team and international teams regarding their use of Agile practices and obstacles they faced in adopting these practices.

In this paper, we present our findings from the international deployment rather than development of the global platform. The aim of the study reported herein is to describe and analyze an industrial case to understand how ASD and DSD can be adopted and used to complement each other in a global environment, and to clarify associated benefits and challenges. Section 3 demonstrates a detailed knowledge of the product development organization, distributed Scrum strategy, process, and the main challenges and practices in deploying Chameleon worldwide.

3 International Deployment of Chameleon: A Case Study

Yahoo! started to adopt and utilize the highly effective and lightweight frameworks of Scrum and Agile in 2005 and has now become one of the largest Agile Enterprise (Benefield 2008). The My Yahoo! global product team also decided to adopt Agile methods in 2006 and the latest version of My Yahoo! ‘Chameleon’ including the international ones, have been developed, localized and released based on the Agile process. This section will give insights into how the global product team and international teams worked together to internationalize and localize the 18 international versions of Chameleon using Agile practices. The teams focused on the benefits rather than on the mechanics of Agile and Scrum.

3.1 Project Background

An easy-to-use custom home page with highly customizable features, My Yahoo! has been a very popular starting point for the web since 1996. This global product has 18 customized international versions with consistent features. The global product team based in Sunnyvale, California and Bangalore, India manage an internationalized platform, while international teams dispersed over three continents manage the localization and configuration as shown in Fig. 1. The local product teams and localization resources in country are allocated as required. As such, the global service group includes offices in Asia Pacific (South Korea, Taiwan, Hong Kong, Singapore, Australia, India and Japan), Europe (Spain, UK, France, Italy and German), Americas (Canada, Brazil, Mexico, and Argentina) and the US.

3.2 Organization Model

The core team in Sunnyvale consists of representatives from each stakeholder group—product management, engineering, QA, marketing, program management, UED (User

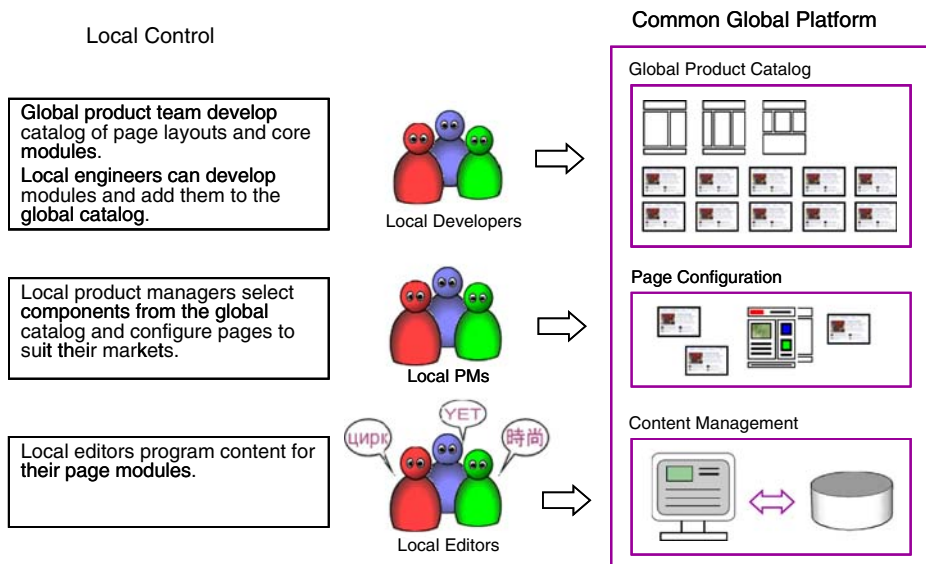


Fig. 1 Chameleon—a global product with local configuration and programming

Experience and Design), CC (Customer Care), architect and international. Each representative is empowered to make decisions on behalf of their group as shown in Fig. 2. The Scrum Master is a new management role introduced by Scrum and project/program managers often take that role. The chief Scrum Master facilitates the whole process and helps people resolve problems in developing the internationalized Chameleon platform while the international Scrum Master facilitates the process of localization and collaboration between the global product team and international teams. Both chief and international Scrum Masters collocate with the global product team in Sunnyvale. The global product team and a few international teams received early coaching and training from the corporate Agile group in 2006, but most international teams were not familiar with Agile methods until 2007 or 2008 (Scotland and Boutin 2008).

Yahoo! Korea and Spain were chosen as double-byte and Latin language representatives and were the first to start the localization of Chameleon among the international teams. The local product managers and engineers traveled to Sunnyvale for the project kickoff and worked closely with the global product team through the international Scrum Master for 2 weeks. The global product team provided up-to-date international deployment manual, checklist, training and coaching as needed. As the internal beta versions of Korean and Spanish Chameleon were released, the next international teams followed a similar track.

3.3 Distributed Scrum Strategy

Sutherland identified three distributed Scrum models commonly observed in practice (Sutherland et al. 2008).

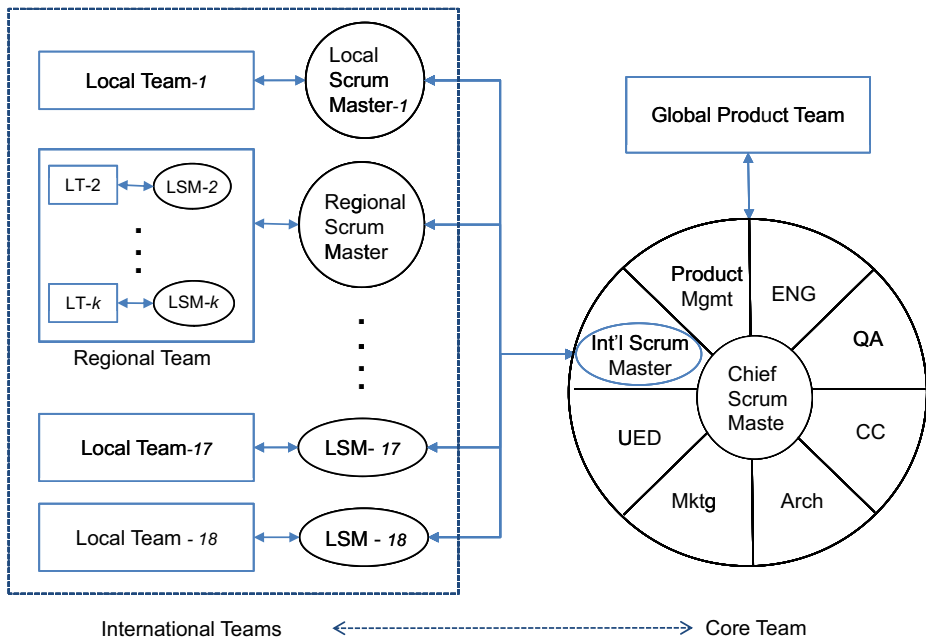


Fig. 2 The global service group for Chameleon

- *Isolated Scrums*: Teams are isolated across multiple locations. In most cases, off-shore teams are not cross-functional and may not be using the Scrum process.
- *Distributed Scrum of Scrums*: Scrum teams are isolated across multiple locations and integrated by a Scrum of Scrums (Sutherland and Schwaber 2007) that meets regularly across locations.
- *Fully distributed Scrums*: Scrum teams are cross-functional with members dispersed across multiple locations.

Isolated Scrums are used mostly where offshore or outsourced teams are not cross-functional and not Agile. These projects often have problems in communication and team relationships. Distributed Scrum of Scrums model is a best practice suggested by the Scrum Alliance. This model partitions work across cross-functional, isolated Scrum teams while eliminating most dependencies between teams. Scrum teams are linked by a Scrum-of-Scrums where Scrum Masters (team leaders/project managers) meet regularly across dispersed locations. The Fully Distributed Scrum model has all teams completely dispersed and each team has members over multiple locations. This model is suggested for experienced Agile teams in multiple locations because cost per story point is the same as it would be for collocated teams (Sutherland et al. 2007, 2008).

In fact, Chameleon globalization project had all three types of Scrums. The global product team dispersed in Sunnyvale and Bangalore; and the European team scattered in UK, France, Spain, German and Italy were implemented based on fully distributed Scrum while the rest of the international teams were implemented based on distributed Scrum of Scrums or isolated Scrum. The international Scrum consisted of 17 local Scrum Masters (local product manager or program manager), 1 regional Scrum Master and an international Scrum Master who is also part of the core Scrum. The global product team was only able to assist 3–7 international teams at the same time due to lack of resources. The dotted boxes in the international Scrum mean “not-active” at that point as shown in Fig. 3.

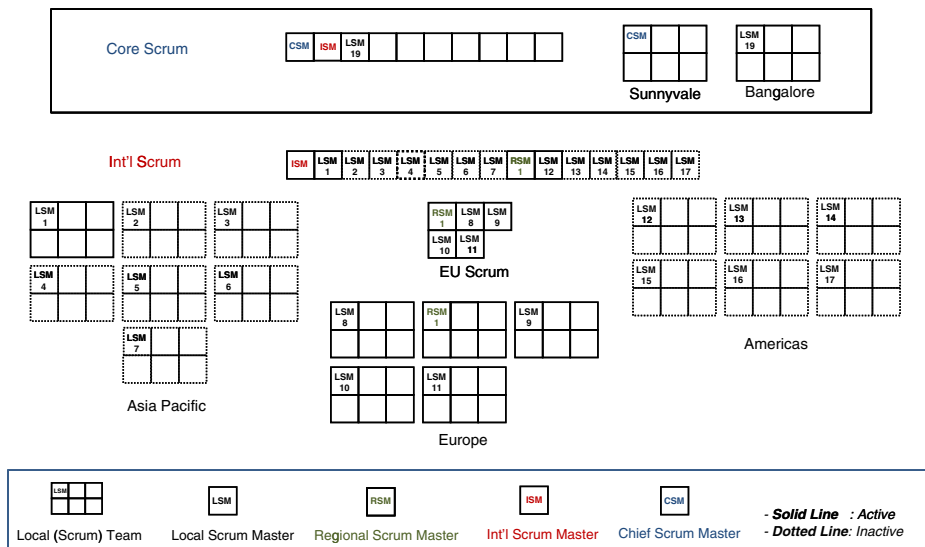


Fig. 3 Complex distributed scrum-of-scrums for international deployment of Chameleon

3.4 Scrum Process in Localization

Localization projects for Chameleon were completed in 2 week length time boxed iterations called “Sprints”. The core Scrum and international Scrums were linked by a Scrum-of-Scrums. The international Scrum Master facilitated cooperation between the global product team and international teams as one of the core team members as shown in Figs. 2 and 3. Their localization work during each sprint was pushed to BETA two times a week and STAGE/PRODUCTION every 2 weeks at the end of every sprint. While the global product team added new features/functions to the platform and fixed platform errors, international teams translated the product from English into the local languages, added local modules, fixed localization errors, and customized features.

Most international teams needed some time called “Sprint Zero”, before kicking off the localization projects. In Sprint Zero, a local team consolidated knowledge and developed product vision & strategy by defining product goals, features/functions, financial objectives, market opportunities, industry trends and competitive differentiators. In subsequent sprints, they planned the next sprint, localized the product, and validated it with users, while iterating as needed as shown in Fig. 4.

3.5 Challenges Faced

In this section we present a set of challenges observed in the course of the adoption process. Our findings are categorized into three areas that global software teams face day-to-day: communication, control and trust (Ramesh et al. 2006).

3.5.1 Communication

There were frequent conference calls and online communications between the global product team and each local team once the localization project kicked off. In addition, the

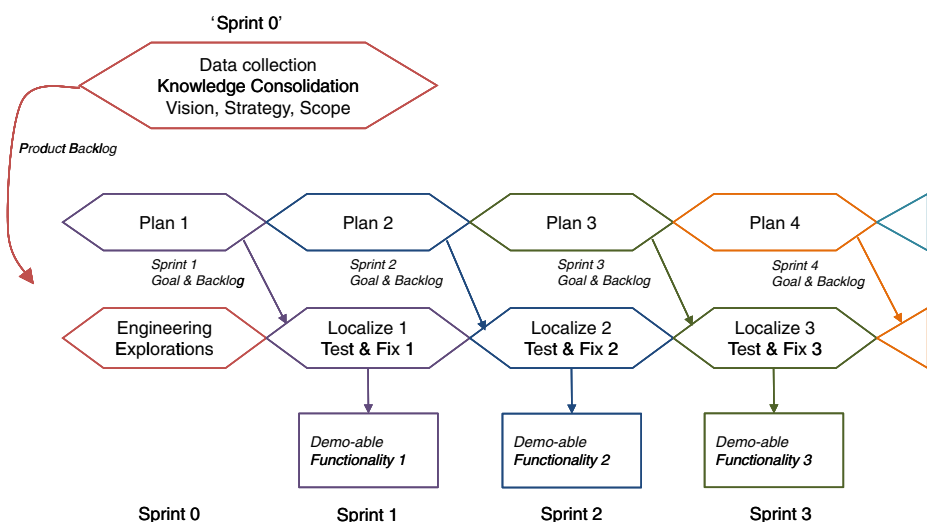


Fig. 4 Scrum process for each international deployment of Chameleon

Sunnyvale team held two daily conference calls over a one month period with the entire international team (8 a.m. and 6 p.m. in Sunnyvale) to cover as many time zones as possible. The Spanish, UK, French, Italian, German, Brazilian, Mexican, Argentinean, Indian teams as well as the teams from Miami and Bangalore attended an early morning call. As well, teams from South Korea, Taiwan, Hong Kong, Singapore, Australia and Japan participated in a late afternoon call. Most of the meetings were conducted by phone and video conferencing. Considerable time was spent clarifying items being discussed, because much of the meaning, tone and emotion were lost through this communication medium. Also the participants often had to be physically at the office to get clear sound and video. On the other hand, the 24/7 work cycle, especially in the US, Bangalore and Asia is a big plus in maximizing productivity. In addition, there were some cultural differences experienced by the global product team in Sunnyvale. They expected all communication lines to be open, however some of international teams were reluctant to discuss negative issues. They only wanted to pass on positive information.

3.5.2 Control

Most local product managers and engineers are responsible for both Front Page and My Yahoo! due to budget constraints. Ironically, My Yahoo!’s biggest rival of localization projects was Front Page because of its higher priority across the organization. Local business conditions and decisions often created disjointed and conflicting priorities within the US product backlog. Lack of awareness of sensitive local issues by the global product team often produced other challenges. One of the characteristics of software globalization projects is that local teams have ownership of their local products, so even though the global product team based in Sunnyvale owns the platform, they have very little control over the local markets or teams.

3.5.3 Trust

In fact, usually the My Yahoo! group doesn’t have a large enough travel budgets allocated for international teams but the international teams try to meet whenever they have international gatherings, such as conferences, corporate training and seminars. They even use (on rare occasions) personal travel during their vacation to meet other teams they have established a relationship with. However, in most of the cases, it is difficult to foster team bonding and collaboration with the distributed teams with few or no face-to-face interaction.

3.6 Practices Used

The global service group implemented a high performance Scrum with XP engineering practices included as shown in Fig. 5. They also devised a set of practices to enable Agile methods to work effectively with globally distributed teams. Even though the entire delivery team never got together to kick off the international deployment projects, productivity and team relationships have been increasing steadily.

3.6.1 Process Flexibility

The global service group used Scrum of Scrums for upper-level coordination of multiple backlogs but didn’t use it by the book. They adopted a “go-local” rule and allowed each team to do their meetings at the most convenient time for them...most of the time. They

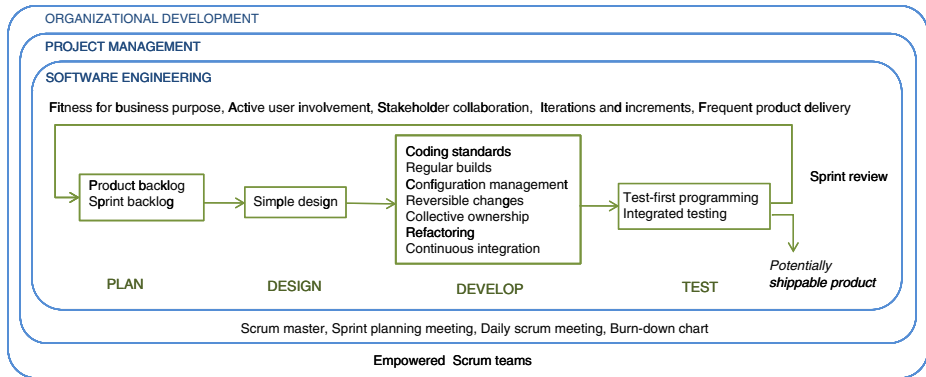


Fig. 5 Scrum and XP Practices used for Chameleon

separated their backlogs according to local/regional customizations and considerations and, they tried to synchronize any local adjustments to sprint backlogs as soon as possible. Also, communication speed was increased by providing less detail at the task level. The Scrum processes were adapted to minimize pain and place value on individuals and interactions over processes and tools.

3.6.2 Empowerment

Each international version of My Yahoo! is an independent product with consistent features based on a single global codebase. Therefore a local team was able to work pretty much independently once they shared the vision/strategy, got proper training/coaching, acquired systems brought up by the global product team based in Sunnyvale and Bangalore. Each local team held Scrum meetings in their own location, and a local Scrum Master emailed a blocks list to the international Scrum Master for their assistance in having the global product team aid in the clearing up process.

3.6.3 Shared Workspaces

The global product team and international teams use the corporate Wiki system to manage the Scrum backlogs, teams, and products. They produce current status and information using simple web page formats and sometimes use photos and/or bios to emphasize the human element. Stakeholders are able to see user stories and backlogs and the status of the project in any desired detail. The corporate Bugzilla and CVS have been used as a defect/task tracking system and software source code control system. The remote teams easily share progress/status and keep shared information constantly updated to maintain interest in activities.

3.6.4 Conferencing

International Scrum Master participated in most core team and global product team Scrum meetings as an international representative, and had a daily progress review with each local Scrum Master. They communicated and collaborated through Telephone, Instant Messaging (IM), Email, Video Conference, WebEx, and informal gatherings or brown-bag lunch talks.

IM is an especially useful tool as a facilitator of communication and helps create a feeling of closeness. The quality of human relationships and communication significantly influences the outcome of a project. Other remote teams have been trying to invest in relationships and communication.

3.6.5 Coaching

The company provided coaching and training in Agile methods to the core team and global product team at the beginning of the project but the benefits were limited to international teams. Instead, international teams collaborated actively through group emails, IMs, Wiki, Bugzilla and Conference calls etc. The teams were encouraged to address problems as they occur and obtain immediate feedback.

4 Empirical Results

We now address our findings from the case study, starting with the discussion of lessons learned from the distributed Scrum strategy and the challenges faced. The second subsection discusses productivity and quality improvements. Subsequently, we discuss why the improvements are so significant.

4.1 Lessons Learned

In this section, we will give a broad overview of distributed Agile issues and challenges, and practices that have been put in place to overcome these challenges. The global service group was implemented based on a core Scrum and an international Scrum, and a centralized management process for international versions of the global product was used as shown in Fig. 3. However, this distributed Scrum strategy would be problematic if simultaneous shipments were required for the 18 countries because the international Scrum Master and global product team would not be able to support all of the international deployment projects at the same time. Also, ideally, international teams should be involved in developing the platform at the beginning of the project in order to add their market requirements to the product. Therefore, we recommend that regional Scrums replace the international Scrum, and regional Scrum Masters become part of the core Scrum as regional representatives. Fig. 6 graphically demonstrates this recommendation for distributed scrum of scrums in a global environment. This would help the global service group utilize international resources more efficiently and effectively and finally enable SimShip (simultaneous shipment) to work.

Distributed teams are a reality in the global software industry. Most challenges faced by distributed teams are related to communication and trust. Each group of practices discussed in Section 3 addresses the challenges faced while incorporating agility in a globally distributed environment. In Fig. 7 we take a broad view of the aforementioned challenges and practices and illustrate how the most relevant practices are applied to each challenge. Mutual respect is a key in building trust. Teams need to trust each other at the beginning of a project and use effective communication to help build a foundation of trust and respect of cultural differences. They also need to understand the importance of getting everyone on the same page as early as possible. Clear communication is compounded by cultural and language differences and it's hard to share tacit knowledge with a globally distributed team. Therefore, it's important to ensure fair and accessible participation for all team members,

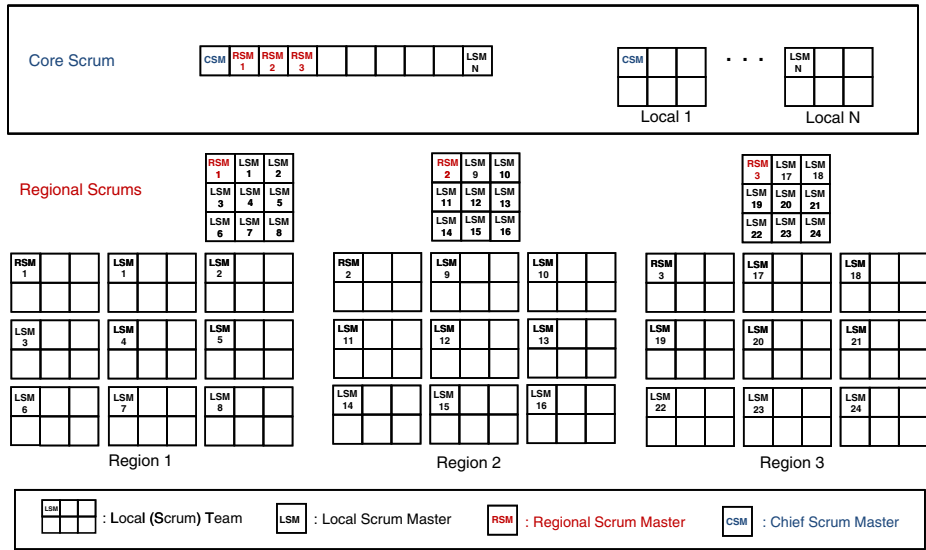


Fig. 6 Suggestions for distributed scrum of scrums for a global environment

maintain a shared vision with all members, and provide transparency of the progress for stakeholders, both inside and outside the organization.

4.2 Productivity and Quality

Distributed Agile approach for the international deployment of Chameleon delivered customer satisfaction and software quality by soliciting customer feedback on a series of

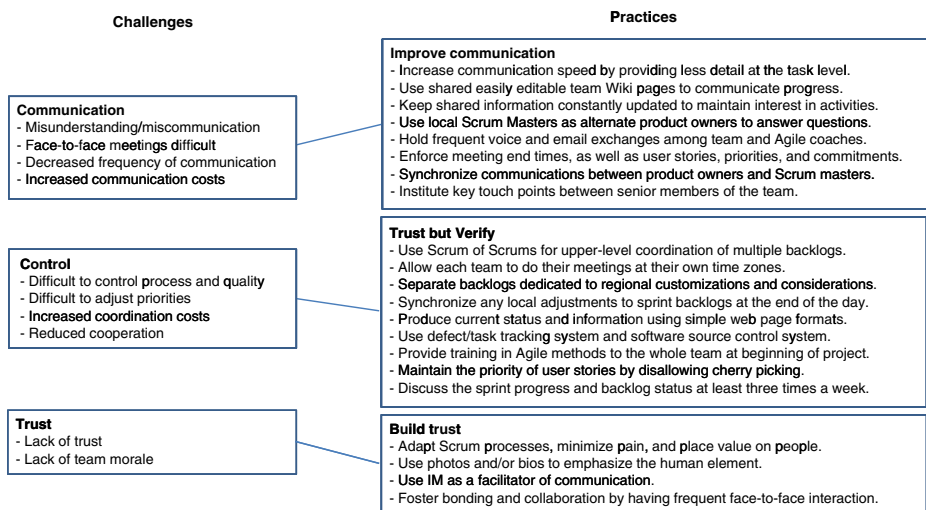


Fig. 7 Mapping challenges to practices

rapid product release. On average, it took 5 months with 3 people from a local team and the global product team to localize/release Chameleon under distributed Agile between 2007 and 2008, while it took 7 months with 3 people from the same distributed team to localize/release Zorro under distributed Waterfall 2 years prior.

My Yahoo! runs on FreeBSD 6.x. The main language is PHP and C++ on the back end, and JS on the front end. The average numbers of completed function points per developer/month and reported showstopper bugs during the localization projects would indicate the productivity and quality of the platform and each process. Therefore, the productivity and quality of international deployment projects of Chameleon have been increased by 40% and 36% respectively, compared to Zorro as shown in Table 1. It may not be proper to compare the old platform to the new one because the localization work scope is somewhat different. Besides, an informal survey of international teams in early 2008 showed about a 30% improvement in the new platform and deployment process. This outcome is consistent with the results of surveys conducted by Yahoo! Agile group (Benefield 2008).

4.3 Discussion

In order to explain the large improvements in productivity and quality of the process in the international deployment of Chameleon, we first identified the three main reasons why the international deployment of Zorro was not successful: 1) Lack of planning and communication between the global product team and local teams; 2) Low priority of the localization project and inexperienced local teams; and 3) Local project manager responsible for multiple functions. We were able to address all of the issues that faced the localization projects with the use of the Agile approach. First of all, to remedy the lack of planning and communication, a Sprint planning and a Scum of Scrums for upper-level coordination were used with the practices mentioned in 3.6.

Second, to address the issue of low priority and inexperienced local teams, Scrum was incorporated into the deployment process as shown in Fig. 4. Scrum mechanisms exist to keep teams functioning at a high level. Iterative releases drive faster ROI and teams feel empowered and enthusiastic as productivity increases. Moreover, Agile engineering practices used for Chameleon (Fig. 5) provided an opportunity for team members to learn from each other by encouraging the team to work closely together.

Third, to tackle the situation where local product/project managers are responsible for multiple functions, a new management role called ‘Scrum Master’ was used. To ensure empowerment of local teams, the international Scrum Master and local Scrum Masters worked diligently on removing impediments, championing agile values, protecting the team from outside interference, and worked as a facilitator and consensus builder.

Table 1 Average localization productivity based on 18 international My Yahoo! (old vs. new version)

My Yahoo! version	Process	Release Period	Person Months	Function Points per dev/month	Showstopper Bugs
Zorro	Distributed Waterfall	2005–2006	21	12.5	108
Chameleon	Distributed Agile	2007–2008	15	17.5	39

5 Conclusions

In today's global environment we are often required to work with widely dispersed teams in all corners of the world in various time zones. In such circumstances it may be more prudent to make regular request for work related information weekly than to submit a huge requirements document and deal with the inevitable delays. When communicating is difficult, we need to shorten the feedback cycle and measure results more frequently.

Agile Principles emphasize face-to-face communication, self-organizing teams, collaboration, changing requirements, frequent delivery, technical excellence and retrospectives etc. Most importantly, face-to-face communication between customers and team members is one of the most critical elements in the Agile Process. Although Agile methods are not easily adaptable to the distributed development environment, there are many IT companies doing Agile distributed software development in various forms (Ågerfalk and Fitzgerald 2006; Lee et al. 2006a, b; Ramesh et al. 2006; Jain 2008), because it has been shown to be effective (Abrahamsson et al. 2002; Cohn 2004; Dybå and Dingsøyr 2008) and has become increasingly popular worldwide (Schwaber 2008; Salo and Abrahamsson 2008). In particular, for software globalization projects to be successful there must be a commitment from the management, product, and development teams (Flarup 2007). Software development is not a defined process, as the fundamental means of development are people. Frequent, unscheduled face-to-face communication breaks down the trust barriers. In addition, unscheduled calls or IMs also provide ample return-on-investment. Once the barriers to trust have been breached through frequent personal interaction, then a project has the potential to be completed successfully.

The purpose of this study is to examine the challenges faced by globally distributed software teams and ascertain the best practices to support project management in such an environment. In this paper, we present distributed Agile as a new approach for managing the international deployment of a global product. The experience gained, and ideas for improvement have come from My Yahoo! and the results are presented in a generalized format in “Suggestions for distributed scrum of scrums for a global environment” (Fig. 6) and “Mapping challenges to practices” (Fig. 7). We hope other organizations reflect on and apply the results herein. They will find that a distributed Agile approach will definitely improve their project quality and productivity, as it has in the fast-paced world of software SimShip.

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References

- Abrahamsson P, Salo O, Ronkainen J, Warsta J (2002) Agile software development methods: review and analysis. VTT Technical report
- Ågerfalk P, Fitzgerald B (2006) Flexible and distributed software processes: old petunias in new bowls. *Commun ACM* 49(10):27–34

- Benefield G (2008) Rolling out agile in a large enterprise, In: Hawaii International Conference on Software Systems (HICSS'41)
- Cohn M (2004) User stories applied: for agile software development. Addison-Wesley
- Dybå T, Dingsøyr T (2008) Empirical studies of agile software development: a systematic review. *Inf Softw Technol* 50(9–10):833–859
- Flarup E (2007) Best practices in software localization. In: Internationalization & Unicode Conference (IUC32), San Jose, California
- Jain N (2008) Distributed agile stage. In: Agile 2008, Available online at <http://www.agile2008.org/stage-distributed.html>. Accessed September 2008
- Lee G, DeLone W, Espinosa JA (2006a) Ambidextrous coping strategies in globally distributed software development projects. *Commun ACM* 49(10):35–40
- Lee O, Banerjee P, Lim KH, Kumar K, Hillegersberg J, Wei KK (2006b) Aligning IT components to achieve agility in globally distributed system development. *Commun ACM* 49(10):48–54
- Ramesh B, Cao L, Mohan K, Xu P (2006) Can distributed software development be agile? *Commun ACM* 49(10):41–46
- Salo O, Abrahamsson P (2008) Agile Methods in European Embedded Development Organizations: a survey study of Extreme Programming and Scrum. *IET Softw* 2:58–64
- Schwaber C (2008) Enterprise Agile Adoption in 2007, Forrester Research
- Scotland K, Boutin A (2008) Integrating scrum with the process framework at Yahoo! Europe. *Agile* 2008:191–195
- Sutherland J, Viktorov A, Blount J, Puntikov N (2007) Distributed Scrum: agile project management with outsourced development teams. In: Hawaii International Conference on Software Systems (HICSS'40), Big Island, Hawaii
- Sutherland J, Schoonheim G, Rustenburg E, Rijk M (2008) Fully distributed scrum: the secret sauce for hyperproductive offshored development teams. *Agile* 2008:339–344
- Sutherland J, Schwaber K (2007) The scrum papers: nuts, bolts, and origins of an agile method. Scrum Inc, Boston



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